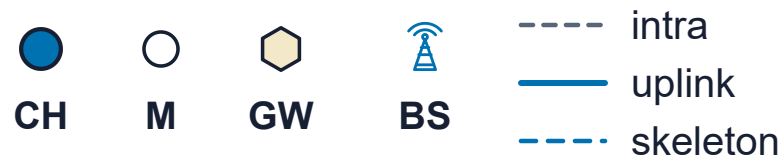
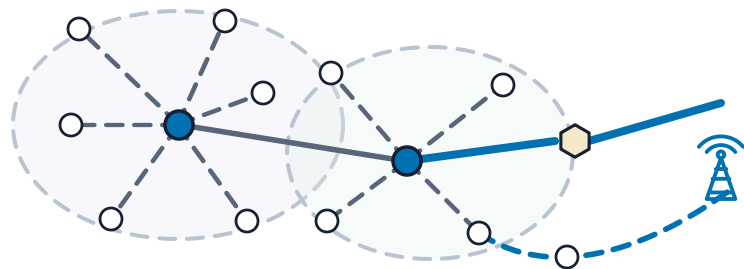
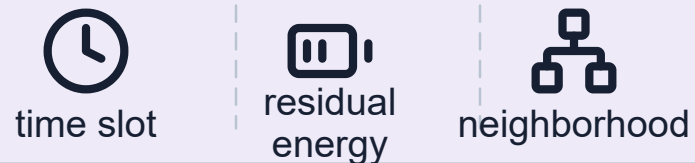




Phase 1 Cluster Formation



1 Round state



2 CH scoring



3 Cluster association



Phase 2 Intra-Cluster Communication

Inputs: energy · link quality · density · buffer

1 CAS mode selector

select one mode

Direct

$M \rightarrow CH$

$O \rightarrow \bullet$

Chain

$M \rightarrow M \rightarrow CH$

$O \rightarrow O \rightarrow \bullet$

Two-hop

$M \rightarrow R \rightarrow CH$

$O \rightarrow \circ \rightarrow \bullet$

O M $\rightarrow$ R $\bullet$ CH

2 Intra-cluster forwarding



3 CH aggregation



Phase 3 Uplink to BS

pre-transmission route selection

